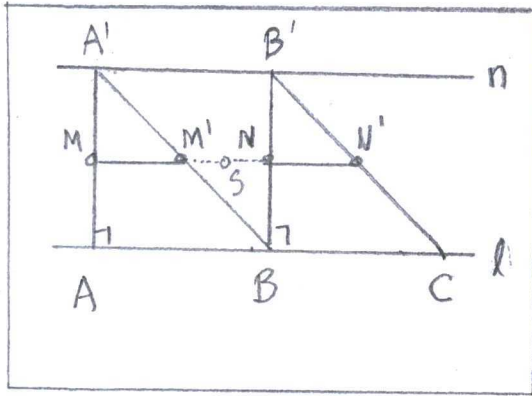


Chapter 6: The Discovery of Non-Euclidean Geometry:

Projects:

1.



(from project) " $\overleftrightarrow{MM'}$ " and " $\overleftrightarrow{NN'}$ " are distinct and intersect at S''

Cite:

Coexter

"Non-Euclidean Geometry"

University of Toronto (1978)

The proof by Coexter supported a non-elementary transformation in hyperbolic, elliptic, and Euclidean spaces. A glide-motion, reflected and rotated in proportions to a point-pair. Coexter also applied Hjelmslev's theorem to Non-Euclidean geometry for a final solution to the proof.

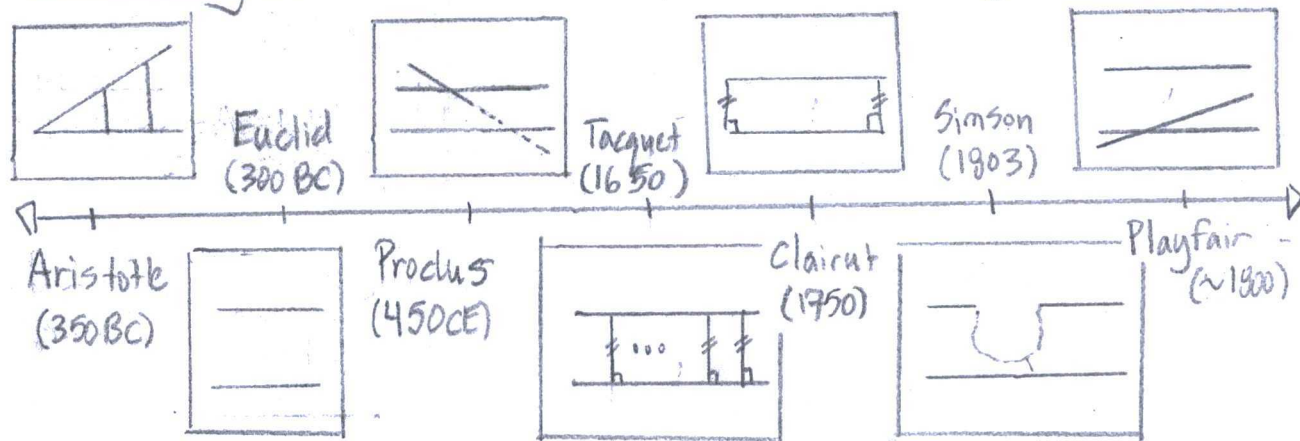
2.

Cite:

Hartshorne, R.

"Geometry: Euclid and Beyond"

University of California, Berkeley (2000)

History of Parallel Postulate, Figuratively:

3. Hilbert's "Foundations in Geometry" had no Appendix III.

4. Hermann von Helmholtz and William Clifford enlightened the English on non-Euclidean geometry in the late nineteenth-century. Multiple papers by Clifford acknowledged: distances, ellipsoids, space-theory, surfaces, and motions "On space-theory of Matter", and

"Mathematical Papers", outlined phenomena.
Clifford speculated, 'geometry is not valid
in curvatures of space,' also, 'refraction
had no results in portions of space.'
The second paper provided equations
to particle's velocity, moment, inertia, momentum,
in space. Both Helmholtz and Clifford
motivated conservation of energy in space.

Cite:

Clifford, W.
"Mathematical Papers"
Cambridge, (1882)

Richards, J.
"Mathematical Visions: The pursuit of
Geometry in Victorian England."
Brown University, (1988)

Note: Chapter 6 related to both Pre-calculus and Calculus 1. Asymptotes were in all subjects. The 'limit' in non-Euclidean geometry was a line or ray toward a point on another line or ray.

Terminology to 'limiting parallels' had two sides: left or right. Calculus described inferior and superior, or outer and inner limits. While, non-Euclidean geometry limited parallels through exterior rays.